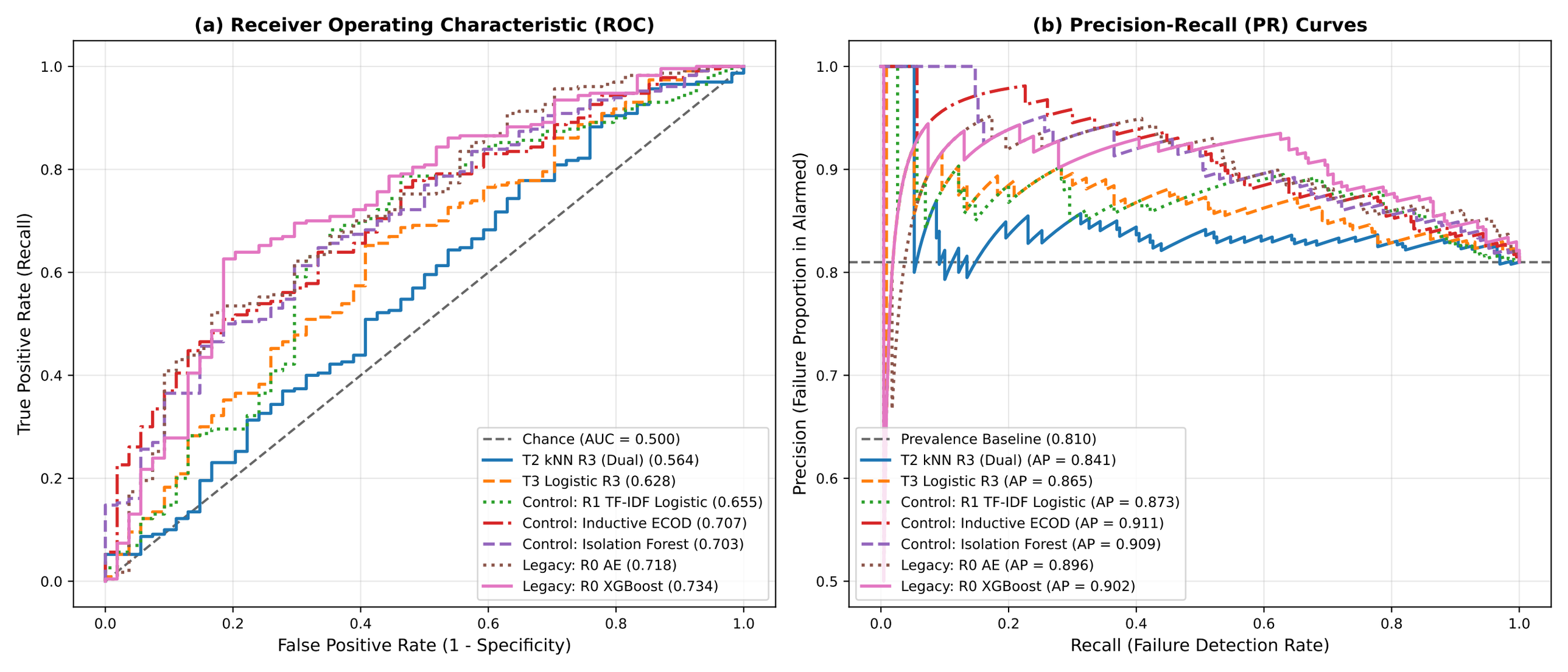
(a) Receiver Operating Characteristic (ROC)
True Positive Rate (Recall)
False Positive Rate (1 - Specificity)
Chance (AUC = 0.500)
T2 kNN R3 (Dual) (0.564)
T3 Logistic R3 (0.628)
Control: R1 TF-IDF Logistic (0.655)
Control: Inductive ECOD (0.707)
Control: Isolation Forest (0.703)
Legacy: R0 AE (0.718)
Legacy: R0 XGBoost (0.734)
(b) Precision-Recall (PR) Curves
Precision (Failure Proportion in Alarmed)
Recall (Failure Detection Rate)
Prevalence Baseline (0.810)
T2 kNN R3 (Dual) (AP = 0.841)
T3 Logistic R3 (AP = 0.865)
Control: R1 TF-IDF Logistic (AP = 0.873)
Control: Inductive ECOD (AP = 0.911)
Control: Isolation Forest (AP = 0.909)
Legacy: R0 AE (AP = 0.896)
Legacy: R0 XGBoost (AP = 0.902)